

Fig. 3: Block diagram of PC-based fingerprint scanner

TABLE II
Pin Details of R305

Pin	Name	Function
1	Vcc	5V power supply
2	Gnd	Signal ground
3	TD	Data output
4	RD	Data input

ers. A CCD is simply an array of light-sensitive diodes called photosites, which generate an electrical signal in response to light photons. Each photosite records a pixel—a tiny dot representing the light that hits that spot. Collectively, light and dark pixels form an image of the scanned scene (a finger, for example). Typically, an analogue-to-digital converter in the scanner system processes the analogue electrical signal to generate a digital representation of this image.

Scanning process starts when you place your finger on the glass plate of R305 module and the CCD camera takes its picture. The scanner has its own light source, typically an array of light-emitting diodes, to illuminate ridges in the fingertip. The CCD system actually generates an inverted image of the fingertip, with darker areas representing more reflected light (ridges in the fingertip) and lighter areas representing less reflected light (valleys between ridges).

Before comparing the fingerprint with the stored data or template, the scanner processor makes sure that the CCD has captured a clear image. It checks the average pixel darkness, or the overall values in a small sample, and rejects the scan if the overall image is too dark or too light.

If the processor finds that the image is crisp and properly exposed, it proceeds to compare the captured fingerprint with fingerprints stored in a file.

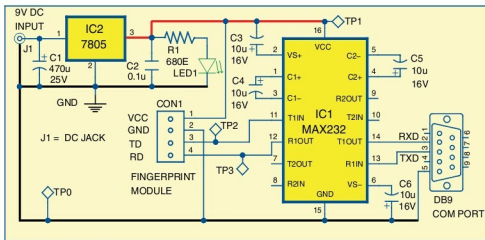


Fig. 4: Circuit of PC-based fingerprint scanner



Fig. 5: Screenshot of program output

Circuit and working

The block diagram of PC-based fingerprint scanner is shown in Fig. 3. The system consists of a fingerprint sensor module that collects data from the finger, MAX232 IC as driver, power supply section and DB9 connector for connecting to the PC.

The circuit of PC-based fingerprint scanner is shown in Fig. 4. The connection setup is simple. All you need is a 4-wire cable for connecting R305 module to the PC through MAX232 circuit. MAX232 is used here because you cannot connect R305 module directly to the PC. The PC is used to process data and display information.

Four connecting terminals are available on R305 module. The functions of each pin are listed in Table II.

The module sends data to the PC through its TD pin and receives data through RD pin. Communication with the PC is done through RS-232 mode, so MAX232 is used between the module and the PC.

Power supply. The circuit is powered by a 9V DC adaptor. The voltage is regulated by a 7805 regulator. LED1

PARTS LIST

Semiconductors:

- IC1 - MAX232 converter
- IC2 - 7805, 5V regulator
- LED1 - 5mm light-emitting diode

Resistors (all ¼-watt, ±5 per cent carbon):

- R1 - 680-ohm

Capacitors:

- C1 - 470µF, 25V electrolytic
- C2 - 0.1µF ceramic
- C3-C6 - 10µF, 16V electrolytic

Miscellaneous:

- DB9 - 9-pin, D-type female COM port connector
- R305 fingerprint module
- 9V-12V DC supply or adaptor

glows to indicate the presence of power in the circuit. Regulated 5V DC powers the MAX232 circuit and R305 module.

Software program

The software is written in 'C#' (C sharp) and compiled on .NET platform using Microsoft Visual Studio 2010. It was tested on a Windows XP machine with .NET Framework 3.5. The screenshot of the program output is shown in Fig. 5.

The software program is used to decode the various image patterns through computer instructions.

R305 module is used to capture your fingerprint image patterns. The captured image is called a live scan. This live scan is digitally processed to create a biometric template (a collection of extracted features), which is stored in a memory and used for matching.

The R305 fingerprint processing involves two steps: enrolling and matching. During enrollment, the user needs to place his finger on the optical sensor twice. The system processes the fingerprint image to generate a template